Arif Mahmud

DevOps Engineer

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Professional Summary

DevOps Engineer with 5+ years of software and infrastructure experience, currently operating production cloud platforms end-to-end on **AWS** and **GCP**. Specialized in container orchestration with **Kubernetes (EKS)** and **ECS**, **Terraform**-driven **GitOps**, and full-stack observability with **OpenTelemetry, OpenObserve and Datadog** — building systems that are automated, secure, and instrumented so issues surface as alerts, not outages. A background as a software engineer informs how I design the platforms products run on.

Skills

- Kubernetes (EKS), ECS & Docker
- ArgoCD, Atlantis & GitOps
- Jenkins & GitHub Actions
- Python & Go
- Terraform / OpenTofu & Ansible
- OpenTelemetry, OpenObserve & Datadog
- Linux Administration & Shell Scripting
- Cloudflare (DNS, CDN & Zero Trust)
- AWS & Google Cloud Platform
- Prometheus, Grafana & Loki
- PostgreSQL, MySQL & BigQuery
- Git & SVN

Work History

DevOps Engineer *02/2024 – Current*

Newroz Technologies Limited — Dhaka

- Own the full infrastructure lifecycle for five production platforms (Hakim, eSIM, Loto, Travluence and internal data pipelines) — from provisioning through steady-state operation on **Amazon EKS, ECS, EC2** and **Docker**.
- Codify infrastructure with **Terraform** and drive **GitOps** delivery through **ArgoCD** and **Atlantis**, giving every infrastructure change a reviewable, PR-based workflow across dev, staging and production.
- Instrument services with **OpenTelemetry** and operate **OpenObserve** and **Datadog** for distributed tracing, log aggregation, alerting and infrastructure monitoring, cutting the time it takes to detect and localize production issues.
- Built EKS monitoring as code: 30-day availability, latency and readiness **SLOs** with multi-window burn-rate paging, 21 Kubernetes and mesh alerts, and 6 dashboards generated in **Python** and applied with **OpenTofu**; cut ~490 noisy metric streams that were overloading the server.
- Automate CI/CD pipelines with **GitHub Actions**, **Jenkins** and **AWS CodeDeploy** for containerized and bare-metal workloads, maintaining disciplined version control with **Git and SVN**.
- Manage domain, DNS and CDN strategy on **Cloudflare** — proxied records, SSL/TLS modes and Zero Trust access policies — alongside **MySQL/PostgreSQL** across cloud and bare-metal environments to sustain high availability.

Software Engineer *04/2021 – 03/2023*

Adecor Business Solutions — Remote

- Built and shipped an e-commerce platform for animation/JSON digital assets end to end, using **Strapi, React** and **Next.js**, deployed on **AWS EC2, S3** and load-balanced **RDS (PostgreSQL)**.
- Integrated **Stripe** for checkout and payment processing across the platform.
- Wrote scalable, maintainable code with long-term stability as a design goal, not an afterthought.

Software Developer *04/2020 – 03/2021*

Spring Rain Ltd — Dhaka, Bangladesh

- Built tour and auth microservices with **Koa.js**, backed by **DynamoDB** and **SQS**, within a broader micro-service architecture.
- Developed the admin back office using **React** and **Redux**.
- Modernized legacy codebases to current development standards, improving functionality and maintainability.

Projects

EKS Platform Monitoring — OpenTelemetry & OpenObserve

- Built an OpenTelemetry agent/gateway pipeline on Amazon EKS (Karpenter on Graviton spot, Istio ambient mesh with Gateway API) into a self-hosted OpenObserve for logs, metrics and traces.
- Defined per-service SLOs (99.5% availability, 95% of requests under 500 ms, 99.9% readiness) measured at the gateway, with burn-rate alerts that page at 14.4× and open a ticket at 6×.
- Shipped 21 platform alerts (crash-loops, NotReady nodes, PVC space, gateway 5xx, mesh certificate expiry, Karpenter failures, dead telemetry) and 6 dashboards covering service map, cluster health, autoscaling and incident triage — all generated in Python and applied with OpenTofu.

Stack: Amazon EKS, Karpenter, Istio Ambient, Gateway API, OpenTelemetry Collector & Operator, OpenObserve, OpenTofu, Python, kube-state-metrics

Enterprise Multi-Account AWS Infrastructure with Terraform & GitOps

- Architected and deployed enterprise-grade AWS infrastructure across 2 environments (staging, production) using Terraform with Atlantis for automated, PR-driven CI/CD workflows.
- Implemented a secure multi-account strategy with AWS IAM Identity Center SSO, IRSA (IAM Roles for Service Accounts), cross-account ECR replication, and automated state management on S3/DynamoDB backends.
- Managed domain routing and edge security with Cloudflare — proxied DNS records, ACM/SSL certificate validation and CDN distribution — in front of the ArgoCD, Grafana and application endpoints.

Stack: Terraform, Atlantis, GitOps, AWS (EKS, VPC, RDS, ECR, S3, Route53, IAM, ACM, EFS, CloudWatch), Helm, ArgoCD, AWS Load Balancer Controller, External DNS, Cloudflare, Grafana, Loki, Prometheus, IAM Identity Center, IRSA, SSL/TLS automation, Secrets Manager

eSIM Platform — Infrastructure & Observability

- Designed and deployed containerized reverse-proxy infrastructure and Cloudflare-managed domain routing for an eSIM management platform across dev and production environments.
- Automated deployment workflows with Docker Compose and AWS CodeDeploy, with OpenTelemetry-instrumented traces shipped to Datadog APM and OpenObserve for logs, alerting and infra monitoring.
- Implemented security best practices including custom headers, request tracing, and environment-specific configuration management.

Stack: Terraform/OpenTofu, AWS (VPC, ALB, RDS, Route53, ACM, ECR, CodeDeploy, CloudWatch), Cloudflare, OpenTelemetry, OpenObserve, Datadog, IaC, CI/CD, Security

Cross-Cloud Data Pipeline — AWS to GCP BigQuery

- Designed and implemented an end-to-end data migration pipeline using Terraform and Go to move data from AWS RDS to Google BigQuery.
- Deployed AWS DMS infrastructure with Terraform for automated database replication across multiple VPC environments.
- Built serverless Cloud Functions in Go with parallel-safe idempotency checks, handling migration control, file operations and batch processing across multiple datasets.
- Added error handling and alerting with migration control tables, lock management and automated cleanup procedures.

Stack: Go, Terraform, Google Cloud Platform (BigQuery, Cloud Functions, Cloud Storage), AWS S3, JSON Schema, IaC

Education

B.Sc., Computer Science & Engineering *12/2018*
Daffodil International University — Dhaka, Bangladesh

HSC *2013*
Government Science College — Dhaka, Bangladesh

SSC *2011*
Manikpur High School — Companiganj, Noakhali

Languages

Bangla — Native / Bilingual

English — Professional Working